

QFT 1

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1 Introduction

1.1 lecture 1.1

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Why study Quantum Field Theory?

QM and SR imply that particle number is not conserved. Taking a box full of particles, QM tells us that there is some uncertainty on the energy: $\Delta E \geq \frac{\hbar c}{L}$. SR tells us that $\Delta E = 2mc^2$. If we reduce the dimensions of the box enough, we can create a particle - antiparticle pair $\Rightarrow$ the number of particles is not conserved.

For this course we will consider $\hbar = c = 1$.

1.1.1 Classical Field Theory

$$\phi(t, \vec{x}) = \phi(x)$$

a field can be thought as a generalization of a collection of generalized coordinates in classical mechanics ($q_a(t) \Rightarrow \infty$ degrees of freedom).

We start by considering the action:

$$S = \int dt L(q_a, \dot{q}_a) = \int dt \underbrace{\int d^3x \mathcal{L}(\phi(x), \partial_\mu \phi(x))}_{L(t)}.$$

where $\mathcal{L}$ is the Lagrangian density.

One example of field is the vector potential.

Scalar Fields

$$\phi(x) \Rightarrow \phi'(x) = \phi(\Lambda^{-1}x)^1$$

¹A scalar field is a function that assigns a single scalar value to each point in spacetime. It can be written as $\phi : R^{1,3} \rightarrow R$, or, more explicitly, $\phi = \phi(x^\mu) = \phi(t, x, y, z)$. Under Lorentz transformations, a scalar field transforms as a scalar, i.e. its value at a given spacetime event is invariant.

under $x^\mu \Rightarrow (x')^\mu = \Lambda^\mu_\nu x^\nu$, where $\Lambda^\mu_\sigma \eta^{\sigma\rho} \Lambda^\nu_\rho = \eta^{\mu\nu}$ and

$$\eta^{\mu\nu} = \eta_{\mu\nu} = \begin{pmatrix} +1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

Simplest theory: Klein-Gordon

$$\mathcal{L}_{KG} = \frac{1}{2} \partial^\mu \phi \partial_\mu \phi - \frac{1}{2} m^2 \phi^2$$

What are the equations of motion?

In classical mechanics we have

$$\frac{\delta S}{\delta q_a(t)} = 0$$

which gives the following equation of motion

$$\frac{\partial L}{\partial q_a} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_a} = 0$$

while in classical field theory we have

$$\frac{\delta S}{\delta \phi_a} = 0$$

which gives the following equation of motion

$$\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} = 0.$$

So, for the KG theory, we can compute the equation of motion

$$m^2 \phi + \partial_\mu \partial^\mu \phi = 0.$$

Noether's Theorem Every continuous symmetry of the action gives rise to a conserved current $j^\mu(x)$. That means $\partial_\mu j^\mu = 0$.

A conserved current implies a conserved charge. To see that we define

$$Q = \int_{\mathcal{R}^3} d^3x \frac{dj^0}{dt} = - \int_{\mathcal{R}^3} d^3x \vec{\nabla} \cdot \vec{j} = 0$$

assuming $j \rightarrow 0$ as $|\vec{x}| \rightarrow \infty$.

This tells us that charges also conserve locally

$$Q_V = \int_V d^3x j^0 \Rightarrow \frac{dQ_V}{dt} = - \int_A \vec{j} \cdot d\vec{S}.$$

For this course, the approach we will use will be to promote the field and its conjugate momentum to operators. This is not the only approach to QFT. For the approach we have chosen we will have to define first the hamiltonian for fields.

Hamiltonian Formalism We start by define the conjugate momentum to the field $\phi_a(x)$ as

$$\pi^a(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}_a}$$

$$\mathcal{H} = \pi^a(x) \dot{\phi}^a(x) - \mathcal{L}(x)$$

where

$$H = \int_{\mathcal{R}^3} \mathcal{H}.$$

We start by considering the lagrangian of the Klein-Gordon field

$$\mathcal{L}_{KG} = \frac{1}{2} \dot{\phi}^2 - \frac{1}{2} (\vec{\nabla} \phi)^2 - \frac{1}{2} m^2 \phi^2.$$

Given $\pi = \dot{\phi}$, we get

$$H = \int d^3x \left(\frac{1}{2} \pi^2 + \frac{1}{2} (\vec{\nabla} \phi)^2 + \frac{1}{2} m^2 \phi^2 \right)$$

which is not manifestly Lorentz invariant.

We want now solve the KG equations and we want to try to make our solutions as Lorentz invariant as possible.

Solution of the KG equation (classical)

$$\partial^2 \phi + m^2 \phi = 0$$

$$\phi(t, \vec{x}) = \int \frac{d^3k}{(\partial\pi)^3} e^{i\vec{k} \cdot \vec{x}} \underbrace{\tilde{\phi}(t, \vec{k})}_{\text{spatial Fourier trans.}}.$$

The solution will be

$$\tilde{\phi}(t, \vec{k}) = A(\vec{k}) e^{-iE_{\vec{k}}t} + B(\vec{k}) e^{-iE_{\vec{k}}t}$$

where $E_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}$.